Electrochemistry

30 questions: 15 multiple choice, 15 structured.

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1 Core Electrolysis [1]

Which statement is the correct definition of electrolysis?

A. The decomposition of an ionic compound, when molten or in aqueous solution, by the passage of an electric current

B. The breakdown of a compound using heat energy alone

C. The reaction of a metal with an acid to produce a salt and hydrogen

D. The process by which a metal ore is converted into the pure metal by heating with carbon

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2 Core Electrolysis [1]

Molten zinc chloride is electrolysed using two graphite rods connected to a battery. What is the name given to the positive electrode?

A. cathode

B. anode

C. electrolyte

D. terminal

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3 Core Electrolysis [1]

Molten lead(II) bromide is electrolysed using inert electrodes. Which pair correctly describes the two products and the electrode at which each forms?

A. lead at the anode, bromine at the cathode

B. hydrogen at the cathode, oxygen at the anode

C. lead at the cathode, bromine at the anode

D. lead at the cathode, oxygen at the anode

4 Core Electrolysis [1]

In the electrolysis of any molten ionic compound, which statement is always true?

A A non-metal element is always produced at the cathode

B Two different metals are produced, one at each electrode

C A metal, or hydrogen, is produced at the cathode, and a non-metal other than hydrogen is produced at the anode

D A metal is produced at the anode and a non-metal at the cathode

5 Core Electrolysis [1]

Dilute sulfuric acid is electrolysed using platinum electrodes. What are the products at the cathode and at the anode respectively?

A hydrogen; oxygen

B oxygen; hydrogen

C sulfur; oxygen

D hydrogen; sulfur dioxide

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6 Core Electrolysis [6]

Molten lead(II) bromide is electrolysed in a school laboratory using two graphite electrodes connected to a power supply.

(a) State which electrode – the anode or the cathode – is the negative electrode, and name the product that forms there. [2]

(b) State the product that forms at the anode. [1]

(c) Describe two observations that would confirm the identity of the product at the anode. [2]

(d) State the name of the molten substance that is decomposed in this experiment. [1]

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7 Core Electrolysis [4]

Molten lithium chloride is held in a crucible and electrolysed using two carbon electrodes connected to a d.c. power supply.

(a) Define the term *electrolysis*. [2]

(b) State the name given to:

(i) the electrode connected to the negative terminal of the power supply [1]

(ii) the molten lithium chloride [1]

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8 Core Electrolysis [1]

Stainless steel cutlery is often electroplated with a thin layer of chromium. What are the main reasons for doing this?

A to improve its appearance and increase its resistance to corrosion

B to make the cutlery lighter and cheaper to produce

C to increase its melting point so it can be sterilised at higher temperatures

D to make the cutlery magnetic

9 Core Electrolysis [5]

A steel spoon is to be electroplated with silver.

(a) State two reasons why the spoon is electroplated with silver. [2]

(b) Describe how the spoon could be electroplated with silver. Your answer should name which electrode the spoon should be, what the other electrode should be made of, and a suitable electrolyte. [3]

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10 Core Electrolysis [1]

Molten calcium bromide is electrolysed using inert electrodes. What are the products formed at the cathode and at the anode respectively?

A bromine; calcium

B calcium; oxygen

C calcium; bromine

D hydrogen; bromine

11 **Extended** Electrolysis [1]

Aqueous copper(II) sulfate is electrolysed using two inert carbon electrodes. What is observed at the cathode?

A a colourless gas is released

B a pink/brown solid metal is deposited

C a green solid metal is deposited

D an orange-brown gas is released

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12 **Extended** Electrolysis [7]

Aqueous copper(II) sulfate is electrolysed twice: once using two inert carbon electrodes, and once, separately, using two copper electrodes.

(a) State the product formed at the cathode. This is the same in both experiments. [1]

(b) With carbon electrodes, state the product formed at the anode. [1]

(c) With copper electrodes, describe what happens at the anode instead. [2]

(d) Explain why, when copper electrodes are used, the blue colour of the copper(II) sulfate solution stays roughly constant, but with carbon electrodes it gradually fades. [3]

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13 **Extended** Electrolysis [1]

Molten sodium chloride is electrolysed using inert electrodes connected to a battery. In the external circuit (the wires connecting the battery to the electrodes), in which direction do electrons flow?

A from the cathode to the anode, through the wires

B electrons do not flow in the external circuit; only ions move

C from the cathode to the anode, through the electrolyte

D from the anode to the cathode, through the wires

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14 **Extended** Electrolysis [6]

Molten magnesium chloride is electrolysed using inert graphite electrodes connected to a battery.

(a) Write the ionic half-equation for the reaction at the cathode. [2]

(b) Write the ionic half-equation for the reaction at the anode. [2]

(c) Describe how charge is carried:

(i) through the electrolyte [1]

(ii) through the external circuit [1]

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15 **Extended** Electrolysis [1]

Concentrated aqueous potassium bromide is electrolysed using inert electrodes. What is formed at the anode?

A oxygen

B hydrogen

C bromine

D potassium

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16 **Extended** Electrolysis [7]

Two separate electrolysis experiments are carried out using inert electrodes: Experiment 1 uses a dilute aqueous solution of sodium iodide; Experiment 2 uses a concentrated aqueous solution of sodium iodide.

(a) State the product formed at the cathode in both experiments, and explain why sodium metal is not formed there. [2]

(b) State the product formed at the anode in Experiment 1 (dilute solution). [1]

(c) State the product formed at the anode in Experiment 2 (concentrated solution). [1]

(d) Explain, in terms of the ions present, why the anode product is different between the two experiments. [3]

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17 **Extended** Electrolysis [1]

Molten potassium chloride is electrolysed using inert electrodes. Which half-equation correctly represents the reaction at the anode?

A $K^+ + e^- \rightarrow K$

B $Cl_2 + 2e^- \rightarrow 2Cl^-$

C $2Cl^- + 2e^- \rightarrow Cl_2$

D $2Cl^- \rightarrow Cl_2 + 2e^-$

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18 **Extended** Electrolysis [6]

(a) Molten lithium bromide is electrolysed using inert electrodes. Write balanced ionic half-equations for the reactions at:

(i) the cathode [1]

(ii) the anode [1]

(b) Dilute sulfuric acid is electrolysed using inert electrodes. Write balanced ionic half-equations, including state symbols, for the reactions at:

(i) the cathode [2]

(ii) the anode [2]

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19 Core Hydrogen–oxygen fuel cells [1]

What is the only chemical product formed when a hydrogen–oxygen fuel cell operates?

A water

B carbon dioxide

C water and carbon dioxide

D hydrogen peroxide

20 Core Hydrogen–oxygen fuel cells [1]

Which two substances react together inside a hydrogen–oxygen fuel cell to generate electricity?

A hydrogen and nitrogen

B hydrogen and oxygen

C methane and oxygen

D hydrogen and carbon dioxide

21 Core Electrolysis [1]

Concentrated aqueous sodium chloride is electrolysed using inert carbon electrodes. What gas is produced at the anode?

A oxygen

B chlorine

C hydrogen

D sodium vapour

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22 Core Electrolysis [1]

Molten strontium chloride is electrolysed using inert electrodes. What products form at the cathode and at the anode respectively?

A chlorine; strontium

B strontium; oxygen

C hydrogen; chlorine

D strontium; chlorine

23 Core Electrolysis [5]

Dilute sulfuric acid is electrolysed in the laboratory using platinum electrodes.

(a) Name the gas produced at the cathode and describe a test that would confirm its identity, including the result. [2]

(b) Name the gas produced at the anode and describe a test that would confirm its identity, including the result. [2]

(c) State the electrolyte used in this experiment. [1]

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24 Core Electrolysis [5]

A car bumper made of steel is electroplated with a thin layer of chromium before sale.

(a) State two reasons why the bumper is electroplated with chromium rather than left as bare steel. [2]

(b) Describe how the bumper could be electroplated with chromium. Your answer should name which electrode the bumper should be, a suitable material for the other electrode, and a suitable electrolyte. [3]

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25 **Extended** Electrolysis [8]

Impure copper is purified using electrolysis. A block of impure copper is used as the anode and a thin strip of pure copper as the cathode. Both are dipped into aqueous copper(II) sulfate.

(a) Write the ionic half-equation for the reaction at the cathode. [2]

(b) Write the ionic half-equation for the reaction at the anode. [2]

(c) Explain what happens to the mass of the anode and the mass of the cathode as electrolysis proceeds. [2]

(d) Explain why the concentration of the copper(II) sulfate solution stays approximately constant during this process. [2]

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26 **Extended** Electrolysis [8]

In industry, concentrated aqueous sodium chloride (brine) is electrolysed continuously using inert electrodes. This is called the chlor-alkali process.

(a) State the products formed at the cathode and at the anode. [2]

(b) Write the ionic half-equation for the reaction at the anode. [2]

(c) Name the solution that remains after electrolysis, and state one industrial use of it. [2]

(d) Explain, in terms of the ions present, why chlorine rather than oxygen is formed at the anode, even though oxygen is normally the preferred product at a carbon anode in dilute aqueous solution. [2]

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27 **Extended** Electrolysis [5]

Dilute sulfuric acid is electrolysed using inert platinum electrodes connected to a battery.

(a) Describe how charge is carried through the dilute sulfuric acid (the electrolyte). [2]

(b) Describe how charge is carried through the wires of the external circuit. [1]

(c) State what happens to hydrogen ions when they reach the cathode. [2]

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28 Core Hydrogen–oxygen fuel cells [4]

A hydrogen–oxygen fuel cell is used to power the electric motor of an experimental car.

(a) State the two gases that are supplied to the fuel cell. [1]

(b) State the only chemical product of the fuel cell reaction, and explain why the car's exhaust does not release carbon dioxide, unlike a petrol-engine car. [3]

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29 **Extended** Hydrogen–oxygen fuel cells [6]

(a) State two advantages of using a hydrogen–oxygen fuel cell, rather than a petrol engine, to power a vehicle. [2]

(b) State two disadvantages of using a hydrogen–oxygen fuel cell, rather than a petrol engine, to power a vehicle. [2]

(c) Suggest one reason why hydrogen fuel cell vehicles are not yet as widely used as petrol-engine vehicles. [2]

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30 **Extended** Hydrogen–oxygen fuel cells [6]

A car manufacturer advertises its hydrogen fuel cell car as "100% pollution-free, with zero emissions."

(a) Explain why this claim is accurate in terms of what comes out of the car's exhaust while it is being driven. [2]

(b) Evaluate whether the claim that the car is "100% pollution-free" is a fair description of its overall environmental impact, considering how the hydrogen fuel might be produced. [4]

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